

Digital Material Passport

Manufacturer	Customer	Business Transaction	
ACME Metal Works GmbH Industrial Park 123 52066 Aachen DE quality@acme-metal.example.com Goods Receiver Global Steel Trading Ltd. - Rotterdam Warehouse Harbor District 45 Pier 7 3089 Rotterdam NL	Global Steel Trading Ltd. Commerce Way 789 2000 Antwerp BE orders@globalsteel.example.com	Order	PO-65478 · 75000 kg Pos. 1-10 · 2025-04-15
		Delivery	DN-98761 · 75000 kg Pos. All · 2025-05-17

Product

Product Name	Structural Steel S355J2+N - Various Shapes	Name (EN)	1.0577
Batch ID	H-79513-03	Marking	S355J2+N laser marked on web surface with clear legibility.
Surface Condition	Hot-rolled	Packaging	Crated bundles of 10 pieces with steel straps in good condition.
Weight	75000 kg	Product Norm	EN 10025-2 (2019)
Production Date	2025-05-16	HS Code	721633 - H sections of iron or non-alloy steel
Country of Origin	DE	CN8 (EU)	72163300 - H-sections of iron or non-alloy steel
Specification	1180-1/ ISO GENERIC - HR - Rev 2024-11-07	HTS (US)	7216330000 - H-sections of iron or nonalloy steel

Chemical Analysis

Heat Number H-79513 - Melting Process BOF+LF - Casting Method ContinuousCasting - Casting Date 2025-05-15 - Sample Location Ladle

Symbol	Unit	Min	Max	Actual	Symbol	Unit	Min	Max	Actual
C	%		0.2	0.17	P	%		0.025	0.017
Mn	%		1.6	1.47	S	%		0.02	0.011
Si	%		0.5	0.25	CEV	%		0.45	0.42

CEV = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.42%

Mechanical Properties

Property	Symbol	Actual	Minimum	Maximum	Method
Tensile Strength ¹	Rm	523 / 525 / 527 MPa - Average 525 - Min 523 - Max 527	470 MPa	630 MPa	EN ISO 6892-1
Yield Strength ¹	ReH	383 / 385 / 387 MPa - Average 385 - Min 383 - Max 387	355 MPa		EN ISO 6892-1
Elongation after fracture ¹	A	22.5 / 23 / 23.5 % - Average 23 - Min 22.5 - Max 23.5	20 %		EN ISO 6892-1
Charpy V-notch Impact Energy 3 specimens tested at -20°C	KV	40 / 42 / 44 J - Average 42 - Min 40 - Max 44 - Std Dev 2	27 J		EN ISO 148-1

¹ 3 specimens tested

Validation

We hereby certify that all material described above has been manufactured and tested in accordance with the requirements of EN 10025-2:2019 and EN 10204:2004 type 3.1. The results comply with the requirements for S355J2+N steel grade.

Validated by Klaus Müller, Quality Control Manager,
Quality Assurance - 2025-05-18